

Stress-Testing AI Governability with LLM-Based Role Simulation

Joseph R. Simons and David A. Broniatowski

¹ Department of Engineering Management and Systems Engineering, The George Washington University, Washington, DC, USA

joseph.simons@gwmail.gwu.edu

² broniatowski@gwu.edu

Abstract. In this paper, we introduce a proof-of-concept LLM-based role simulation protocol for stress-testing whether an AI governance framework adds value. Using the NIST AI Risk Management Framework (RMF) as the test object, we examine AI-supported lending using a 4 (organizational role: enterprise executive, line executive, middle manager, system owner) $\times$ 2 (deployment case: bounded ML underwriting-risk model vs. workflow-embedded LLM underwriting copilot) $\times$ 3 (governance stressor: weak justification, apparent disparity, and workflow drift/overreliance) design, with five replications per cell. Results suggest that actors could usually interpret the RMF from their assigned role and identify relevant delegation pathways. The harder problem was translating role-level activity into cross-level, authority-backed governance value, especially when the governed object was a workflow-embedded AI system-in-use rather than a bounded model. The paper contributes a diagnostic approach for evaluating whether AI governance frameworks help organizations implement meaningful governance rather than merely producing compliance artifacts.

Keywords: AI governance · NIST AI RMF · trustworthy AI · sociotechnical systems · role simulation

1 Introduction and Framing

The NIST Artificial Intelligence Risk Management Framework (AI RMF) presents a useful puzzle. It is a practical, widely recognized, and explicitly sociotechnical framework for helping organizations manage AI risks and promote trustworthiness and responsible AI [6]. Yet the practical difficulty of AI governance does not disappear once a framework names the right risks, identifies trustworthiness characteristics, or specifies governance activities. A framework can be known, endorsed, and apparently useful while still failing to become organizational governance, as has been observed in studies of AI RMF adoption [4].

This paper starts from that gap. Trustworthy AI is often discussed through desirable system characteristics such as validity, reliability, safety, fairness, and related properties. These characteristics matter, but in consequential organizational settings they do not become trustworthiness by themselves. An AI system

can be documented, monitored, assessed, and reviewed while still shaping organizational judgment in ways that are difficult to observe, justify, or correct. The practical question is therefore not only whether an AI system has trustworthy characteristics, but whether the organization can govern the AI system-in-use.

This is a question of where the system boundary is drawn. If the boundary for trustworthy AI is drawn around the model or technical AI system, governance tends to take an audit-and-assurance form: evaluating controls, performance, and artifacts against specified requirements. This risks treating sociotechnical AI harms within too narrow a system boundary [8]. If the boundary is drawn around the organization that owns, deploys, relies on, oversees, and corrects the AI system-in-use, the problem looks different. Recent AI safety work makes a similar point: AI cannot be safeguarded through model- or algorithm-level design choices alone, but requires attention to context of use, affected stakeholders, and the institutional environment in which the system operates [3]. Trustworthiness also depends on whether organizational actors can interpret what the system is doing, determine how its outputs should matter, bound acceptable reliance, detect deviations, escalate concerns, and authorize correction.

We use *governance value* to distinguish governance that improves organizational control and decision quality from governance that merely produces activity or artifacts. Governance has value when it helps actors see what is happening in the system-in-use, interpret why it matters, determine whether reliance remains justified, coordinate across roles, and authorize correction. Documentation, dashboards, controls, reviews, and other artifacts may support this work, but they become valuable only when they improve the organization’s ability to see, interpret, decide, and correct.

The RMF aspires to more than governing reliance. It aims to support trustworthy and responsible AI across multiple characteristics and lifecycle functions. In this study, however, reliance provides a minimal diagnostic test. Lee and See’s account of trust in automation treats the central design problem as one of appropriate reliance: users should neither over-rely on automation nor fail to use it when reliance is warranted [5]. We adapt that concern from individual human-automation interaction to organizational AI governance. Here, reliance means the extent to which an organization allows an AI system or its outputs to materially shape consequential human judgment, workflow, or final decisions. If a governance framework cannot help an organization make reliance justified, bounded, observable, and correctable, it is unlikely to support trustworthy AI governance in any stronger sense.

The analysis uses two concepts to evaluate whether RMF-guided activity becomes governance value. *Governance translatability* asks whether framework language can become role-usable, cross-level governance rather than local compliance activity. The issue is whether local risk signals can reach actors with authority and return as decisions that change practice. *Structural fit* asks whether the framework’s implied object of governance matches the system it is being asked to govern. The issue is whether RMF categories can attach to the actual

AI system-in-use: its purpose, outputs, workflow, reliance patterns, risk signals, authority pathways, and corrective mechanisms.

To examine this problem, the paper introduces a proof-of-concept Large Language Model (LLM)-based role simulation protocol and applies it to the NIST AI RMF in AI-supported lending. The $4 \times 2 \times 3$ design, with five replications per cell, varies four organizational roles, two deployment cases, and three governance stressors: weak justification, apparent disparity, and workflow drift or overreliance. The comparison varies the object of governance; it does not classify all Machine Learning (ML) or LLM systems. The ML underwriting-risk model has comparatively stable outputs, signals, and control points, including scores, thresholds, validation results, adverse-action reasons, and disparity indicators. The LLM underwriting copilot is more workflow-embedded and interpretive: its outputs may shape salience, narrative framing, rationale construction, documentation practice, and reliance.

The paper makes three contributions. First, it reframes AI governance framework adoption as a sociotechnical governability problem: the issue is not merely whether a framework can be implemented, but whether framework-guided activity becomes governance value across organizational levels. Second, it contributes a diagnostic protocol for stress-testing whether an AI governance framework creates role-usable, cross-level, authority-backed governance value. Third, it shows that framework translation depends on organizational translatability: whether local risk signals can move to actors with authority, and whether their decisions can move back down as changed practice. It also depends on the structural fit between framework categories and the governed object. The broader implication is that AI governance frameworks should be evaluated not only by whether they define trustworthy characteristics or produce governance artifacts, but by whether they make AI systems-in-use governable. That requires existing linkages that allow movement between organizational levels of abstraction [7].

2 Protocol and Study Design

We use LLM-based role simulation as a structured analytic probe to test whether RMF language can support coherent, role-positioned governance reasoning under controlled variation. The approach builds on recent work using language models to simulate human samples and experimental responses [2, 1]. The protocol uses a $4 \times 2 \times 3$ design with five replications per cell: four organizational roles, two AI-supported lending deployments, and three governance stressors, producing 120 role-positioned responses. The roles were specified through templates defining each actor’s function, authority, responsibilities, informational vantage point, baseline competence, and value orientation. The design is summarized in Table 1. The two deployment cases vary the governed object. The ML case is a traditional underwriting-risk model that scores or flags loan applications for elevated credit risk and has comparatively stable governance control points, including thresholds, validation results, drift metrics, adverse-action reasons, and disparity signals. The LLM case is an underwriting copilot that reviews appli-

Table 1. Study design

Component	Values
Roles	Enterprise executive; line executive; middle manager; system owner/operator
Deployments	ML underwriting-risk model; LLM underwriting copilot
Stressors	Weak justification; apparent disparity; workflow drift/overreliance
Simulations	$4 \times 2 \times 3$ with 5 replications per cell = 120 role-positioned responses

cant materials, summarizes risk factors, surfaces inconsistencies, and supports credit-risk assessment through narrative or semi-structured outputs.

The three governance stressors test different governability problems: weak justification, apparent disparity, and workflow drift or overreliance. The weak-justification stressor involves a borderline adverse outcome supported by thin, generic, or unclear justification. The apparent-disparity stressor involves a disparity signal across applicant groups whose source and accountability pathway are unclear. The workflow-drift or overreliance stressor involves human decision makers treating AI output as increasingly dispositive or allowing AI-generated framing to displace independent judgment.

Each simulation assigned the LLM to one role, one deployment case, and one stressor, and instructed it to reason only from the assigned role using the NIST AI RMF. The prompt required structured role-positioned output covering role interpretation, RMF-supported governance arrangements, authority and escalation, reliance and assurance, trust-related risks, and limits. This set-up created an idealized, best-case test of governance translation rather than an unscaffolded test of role recognition or a prediction of how untrained employees would reason in practice.

The analysis proceeded in two steps. First, each response was assessed for whether the RMF could be translated into value-adding governance for the assigned role, deployment case, and stressor. Second, where governance translation was present, the response was assessed for whether that governance plausibly reduced relevant trust-related risks. This second judgment considered both consumer-facing risk, including unfair, biased, opaque, or otherwise harmful outcomes for applicants, and organizational risk, including legal, regulatory, operational, and reputational exposure. It also asked whether the response made reliance on the AI system-in-use more justified, bounded, observable, and correctable. Each dimension was scored on a 0–2 scale, corresponding to weak, partial, or strong evidence of the relevant capability. Governance translatability was assessed through three subdimensions: role translation, cross-level governance value, and authority connection. These ask whether the assigned role can use the RMF, whether that work matters beyond the local role, and whether it connects to authority and correction. The subdimensions were scored on the same 0–2 scale and averaged into the governance translatability score. Structural

Table 2. Diagnostic dimensions

Dimension	Diagnostic question
Actor-level interpretive fit	Can the actor make sense of the problem, its role relevance, and its expected contribution from the assigned role?
Governance translatability	Can RMF language become role-usable, cross-level governance rather than local compliance activity?
Delegation legibility	Can authority, accountability, consultation, escalation, and execution responsibilities be plausibly mapped?
Structural fit	Does the RMF’s implied object of governance match the AI system-in-use it is being asked to govern?

fit was assessed through meaning determinacy, closure, and boundary containment. These ask whether the system’s purpose, role, acceptable behavior, and performance criteria are stable enough for governance; whether problems can be interpreted and corrected without reopening basic questions about intended role, actual use, workflow, incentives, authority, or evaluation criteria; and whether relevant risks, harms, corrective mechanisms, and governing authority remain within the system-of-concern. To reduce scoring inflation, full credit required explicit textual evidence connecting RMF-guided activity to decisions, escalation, reliance, correction, or organizational control; generic references to controls, documentation, dashboards, reviews, or escalation were insufficient on their own.

3 Results

Three findings organize where RMF-guided reasoning translated into governance value and where that translation remained partial. These results are summarized in Table 3.

3.1 Local Interpretation and Delegation Were Not the Bottleneck

Actor-level interpretive fit and delegation legibility were constant across the scaled run. This means the simulated actors could generally interpret the RMF from their assigned organizational level and identify relevant authority, accountability, and escalation relationships. The result should be read in light of the structured prompt, which gave each actor a defined role and required role-positioned governance reasoning. Under those conditions, the limiting problem was not whether actors knew what mattered or who had formal responsibility. It was whether role-level activity could become cross-level, authority-backed governance value.

Table 3. Governance translatability and structural fit by role and deployment

<i>Panel A. Score by role and deployment (scores were consistent across runs)</i>						
Role	ML GT	ML SF	LLM GT	LLM SF		
EE	2.00	2.00	2.00	1.00		
LE	2.00	2.00	2.00	1.00		
MM	1.67	2.00	1.67	1.00		
SO	1.67	2.00	1.67	1.00		

<i>Panel B. Step judgments by role and deployment</i>						
Role	ML S1	ML S2	ML Value	LLM S1	LLM S2	LLM Value
EE	15 yes	15 yes	15 visible	15 yes	15 partly	15 visible
LE	15 yes	15 yes	15 visible	15 yes	15 partly	15 visible
MM	15 partly	15 partly	15 mixed	15 partly	15 partly	15 mixed
SO	15 partly	15 partly	15 mixed	15 partly	15 partly	15 mixed

Note. Each role-by-deployment entry summarizes 15 responses: three stressors with five replications each. Scores use the 0–2 scale in Table 2. GT = governance translatability; SF = structural fit; S1 = Step 1; S2 = Step 2. EE = enterprise executive; LE = line executive; MM = middle manager; SO = system owner.

3.2 Governance Translatability Varied by Role

Governance translatability varied by organizational role. Enterprise executive and line executive responses translated RMF-guided activity most cleanly because those roles could connect risk signals to risk appetite, operating limits, resource allocation, escalation expectations, and decisions to continue, restrict, remediate, or suspend use. Middle-manager and system-owner responses were still role-appropriate, but their value was more conditional. They could produce procedures, monitoring routines, evidence, telemetry, issue logs, and escalation triggers, but those mechanisms depended on higher-level authority or cross-functional follow-through.

This shows the difference between local governance activity and organizational governance value. A role can know what to do and identify who should act, while still lacking the authority needed to move the needle. Governance value appears when local signals travel to actors with authority and return as changed practice.

The central bottleneck was whether there was a path from signal detection to interpretation, escalation, decision quality, and authorized correction. The strongest responses identified the risk signal, explained why it mattered from the assigned role, specified who needed to know, clarified what authority was required, and described corrective options. Where responses stayed at the level of local activity, procedural review, or generalized oversight, governance value remained partial. Where responses linked detection, interpretation, escalation, executive decision quality, and correction, the RMF was more plausibly translated into governable reliance.

3.3 Structural Fit Varied by Deployment

Structural fit varied by deployment. The ML underwriting model was easier to govern because the objects of governance are model-centric: scores, thresholds, validation results, override behavior, adverse-action reasons, disparity measures, and drift metrics. These did not guarantee good governance, but they gave actors familiar signals and correction pathways.

The LLM underwriting copilot was harder to govern because the relevant object became the AI system-in-use. Risk emerged through workflow, narrative framing, copied rationales, documentation practice, user reliance, and underwriter judgment. These risks were harder to observe and harder to correct through ordinary model-governance mechanisms. The stressors exposed different versions of this problem, but they did not drive the main pattern; they functioned as probes of whether local signals could become authority-backed correction under different kinds of governance pressure.

4 Discussion and Conclusion

Results show that the primary differences were in governance translatability and structural fit. Actors could generally interpret the RMF from their assigned role and identify relevant delegation pathways. The harder organizational problem was whether role-level activity could become cross-level, authority-backed governance value. Governance activity can originate at any organizational level. The finding is that it became more organizationally consequential, and more likely to add value or reduce risk, when it connected to decision authority, escalation, and correction. The structural-fit problem was different. In the ML case, RMF governance was easier because risk could be connected to stable outputs, thresholds, validation results, and disparity signals. In the LLM case, governance was more fragile because risk emerged through workflow, narrative framing, documentation practice, and reliance.

This matters because some precursors of trustworthy AI reside in governance capability, not in the model or assurance artifact alone. The RMF was most useful when actors translated framework language into mechanisms that moved risk signals to authority and returned as correction. Documentation, dashboards, controls, and review routines became governance value only when they helped the organization see what was happening, determine whether reliance remained justified, and authorize change. The ML and LLM contrast reinforces the point: governance was more legible when the governed object had stable control points, and less legible when risk emerged through workflow, documentation practice, and reliance.

The practical implication is that AI governance-framework adoption should be treated as capability building, not only implementation. Organizations need activity-to-decision mapping, signal-to-authority pathways, role-specific value propositions, and structural-fit assessment. They should ask whether governance activities move information to actors with authority, return decisions to practice, and make the AI system-in-use more observable and correctable.

This study is deliberately bounded. Because the prompt supplied role profiles and required use of the NIST AI RMF, the results should be read as an idealized test of governance translation rather than a prediction of how practitioners would reason without those supports. The cases are stylized and the scoring is qualitative. Future work should compare LLM-generated responses with human role-positioned exercises, use multiple raters, test additional domains and frameworks, and examine team-level governance reasoning.

Despite these limits, the protocol provides a useful diagnostic. It identifies when a framework becomes governance value, when it remains local compliance activity, and when the governed object strains the framework’s implied system model. For consequential AI systems, trustworthy AI depends not only on models or artifacts, but on whether the organization can bound, observe, interpret, and correct reliance on the AI system-in-use. AI governance frameworks should therefore be judged by whether they help organizations build sociotechnical governability.

Acknowledgements. This work was supported by the NIST-NSF Institute for Trustworthy AI in Law and Society (TRAILS), under Award No. 2229885. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the NSF or NIST.

References

1. Aher, G., Arriaga, R.I., Kalai, A.T.: Using large language models to simulate multiple humans and replicate human subject studies. In: Proceedings of the 40th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 202. PMLR (2023)
2. Argyle, L.P., Busby, E.C., Fulda, N., Gubler, J.R., Rytting, C., Wingate, D.: Out of one, many: Using language models to simulate human samples. *Political Analysis* **31**(3), 337–351 (2023). <https://doi.org/10.1017/pan.2023.2>
3. Dobbe, R.I.J.: System safety and artificial intelligence. In: The Oxford Handbook of AI Governance (2022)
4. Kolo, F.H.: From framework to practice: Barriers and enablers to rmf adoption in mid-sized enterprises. *Asian Journal of Research in Computer Science* **18**(5), 459–479 (2025)
5. Lee, J.D., See, K.A.: Trust in automation: Designing for appropriate reliance. *Human Factors* **46**(1), 50–80 (2004)
6. National Institute of Standards and Technology: Artificial intelligence risk management framework (ai rmf 1.0). Tech. Rep. NIST AI 100-1, National Institute of Standards and Technology (2023). <https://doi.org/10.6028/NIST.AI.100-1>
7. Rasmussen, J.: The role of hierarchical knowledge representation in decisionmaking and system management. *IEEE Transactions on Systems, Man, and Cybernetics SMC-15*(2), 234–243 (Mar/Apr 1985)
8. Selbst, A.D., Boyd, D., Friedler, S.A., Venkatasubramanian, S., Vertesi, J.: Fairness and abstraction in sociotechnical systems. In: Proceedings of the Conference on Fairness, Accountability, and Transparency. pp. 59–68. Association for Computing Machinery (2019). <https://doi.org/10.1145/3287560.3287598>